ILONA DEMLER

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Education

California Institute of Technology — Pasadena, CA
Ph.D. Student in Computing and Mathematical Sciences — Sept. 2023 – Present
Computational Vision Lab
Advisor: Pietro Perona

Harvard University — Cambridge, MA
B.A. in Physics — Sept. 2018 – May 2023
Thesis Advisor: Michael Brenner

Thesis: "Pose From Home: Running Clinical Trials for Neurodegenerative Diseases From Patients' Homes Using Compressed Human Pose Estimation Models on TinyML Devices"

Publications

- "CalTennis: Large Multi-View Tennis Video Dataset and Benchmark of 3D Human Pose"
  Demler, I., Xie, X., Werner, B., Szczuka, A., Perona, P. (2026). *Under review.* [project] [dataset]
- "TaskNPoint: How to Teach Your Humanoid to Hit a Backhand in Minutes"
  Werner, B., **Demler, I.**, Perona, P., Ames, A. D. (2026). *Under review.* [project] [code]
- "ITTO: Is This Tracker On? A Benchmark Protocol for Dynamic Tracking"
  Demler, I., Chauhan, S., Gkioxari, G. (2025). *NeurIPS.* [project] [dataset]

Awards & Fellowships

- 1st Place at MOLTEN Spatial AI Hackathon, a16z LA Tech Week — 2026
- Outstanding Mentor, Caltech SURF Undergraduate Fellowship Program — 2026
- NSF Graduate Research Fellowship — 2025-2028
- Caltech EAS Chair Scholar — 2023-2025
- Harvard Physics Department Carol Davis Prize — 2023
- Kernel Fellowship — 2022
- PRISE Fellowship — 2020
- John Harvard Scholar — 2019

Work Experience

Brenner Lab, Harvard University — Cambridge, MA
Undergraduate Researcher — May 2022 – Aug. 2023

- Prototyped the first device that runs continuous pose detection, computes biometrics, and stores data in cloud server to preserving patient privacy; device was used in funded neuro-degenerative treatment trial.

Disney Research — Zurich, Switzerland
Intern, Machine Learning and Data Intelligence Group — June – Aug. 2022

- Implemented custom PyTorch autograd and backprop modules for privacy-preserving model training.

DreamsAI Cambridge, UK
Software Engineering Intern Apr – Aug. 2021

- Designed and built cryptocurrency holding platform using a hierarchical deterministic wallet setup compatible with Ethereum and Binance Smart chains.
- Led team of three software engineers developing horserace betting model. Improved web-scraping and feature engineering pipeline, boosting profits by 1.2x and data saving speed by 2x.

Acronis Remote
AI Intern Jan. – Apr. 2021

- Built demo to calculate relevant joint and motion metrics including shot outcomes by detecting and analyzing basketball free throws. Built a data labeling pipeline in CVAT and fine-tuned Detectron2 and OpenPose models. Showcased results to a partner NBA team.

Cadence Design Systems Remote
Software Engineering Intern May – Dec. 2020

- Modeled SPICE circuit outputs, reducing simulation time from 1 week to <2 hrs. Used Ray library for parallelization and parameter tuning, presenting results at quarterly R&D teams meeting.

Selected Talks

- **Meta Sensors Workshop, Caltech**. "RallyAI: A Challenge for Physical Intelligence" 2026
- **Novartis Data Science Group**. "Privacy-preserving pose-analysis for at-home clinical trials" 2023
- **IAIFI Summer School**. "Privacy-preserving pose-analysis device for at-home clinical trials" 2023

Service & Mentorship

- **Reviewer for International Conferences:** NeurIPS 2026
- **Research mentor:** Caltech SURF Fellowship Program 2026

Skills

Programming: Python (NumPy, SciPy, Pandas, Matplotlib), Git, Jupyter.
ML & Vision: PyTorch, SMPL-X; W&B, TensorBoard; Transformers, DiTs, SSMs; reinforcement learning.
Hardware: NVIDIA Jetson, Google Coral Dev Board, Meta Quest VR Headset, motion capture.
Data Annotation: CVAT, built custom 3D annotation tools, annotator management on Upwork and Amazon Mechanical Turk, Hugging Face Datasets.
3D & Capture: MuJoCo, Isaac Sim, spatial and audio-based sensor synchronization.
Tools: Git, Jupyter, Matplotlib, Pandas, SciPy.
Languages: English (native), Russian (native).

Extracurriculars

Ballet

- Trained at Bolshoi Ballet Academy (Moscow)
- World Ballet Competition finalist
- Youth America Grand Prix top 24; offer to Boston Ballet Pre-Professional Program
- Harvard Modern Dance Company Member

Tennis

- High school varsity team captain and league MVP.